

FACULTY	Agriculture, Engineering and Natural Sciences		
SCHOOL	Engineering and the Built Environment		
DEPARTMENT	Mechanical and Metallurgical Engineering		
SUBJECT	ENGINEERING MECHANICS II		
SUBJECT CODE	I3641NM		
DATE	17 JUNE 2025		
DURATION	2 1/2 Hours	MARKS	100

First Opportunity Examination

Examiner: Prof. I. O. Ohijeagbon

Internal Moderator: Dr. E. Shaanika

External Moderator: Prof. M. Waheed (FUNAAB-NIGERIA)

This question paper consists of **4 pages** including this front page.

Instructions

1. Closed book examination
2. Read the questions carefully
3. This question paper consists of **Five** questions
4. Answer **any 4 Questions**
5. Each question carries **25 marks**
6. Neatly drawn sketches should be used where required

Question One**[25 Marks]**

(a) A car begins its motion from rest and accelerates uniformly. After covering a distance of 200 meters, it attains a velocity of 15 m/s. Determine:

- (i) The magnitude of the car's acceleration.
- (ii) The duration of time taken to travel the 200-meter distance.

(4 Marks)

(b) Referring to the car in part (a), suppose it is now moving at a maximum velocity of 35 m/s when the driver applies the brakes. The car's speed decreases from 35 m/s to 10 m/s for 15 s, then from 10 m/s to 0 m/s over the next 10 seconds. Determine:

- (i) The constant deceleration during each of these two time intervals.
- (ii) The overall average deceleration from the moment the brakes were applied until the car comes to a complete stop.
- (iii) The car's average speed over the entire journey from when it starts from rest as stated in part (a), accelerates to its maximum speed, and continues until it comes to a complete stop.

(21 Marks)**Question Two****[25 Marks]**

(a) A study was conducted to investigate the curvilinear motion of a projectile. Assuming the projectile undergoes constant acceleration, outline the analytical techniques employed to characterize the projectile's kinematics with an appropriate illustrative diagram. (8 Marks)

(b) A ball is kicked from point A with an initial velocity of $v_A = 10 \text{ m/s}$ at an angle of 30° to the horizontal, as shown in Figure Q. 2(b). Determine the following parameters for the ball's projectile motion:

- (i) The maximum height h .
- (ii) The total time of flight.
- (iii) The horizontal range A to C.
- (iv) The speed of the ball when it strikes the ground.
- (v) The maximum height and total time of flight if the ball is kicked at an angle of 60° .

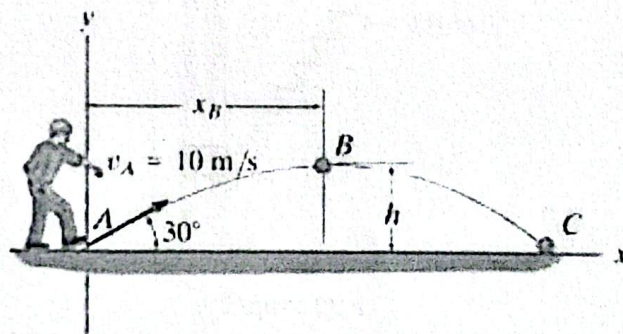
(17 Marks)

Figure Q. 2(b)

Question Three**[25 Marks]**

(a) A 4 kg particle moves in 3D space under the influence of three force components:

$F_x = 12 \text{ N}$, $F_y = -8 \text{ N}$, $F_z = 20 \text{ N}$. Calculate:

(i) The acceleration vector $\bar{a}$ of the particle

(ii) The magnitude and unit vector of the acceleration $\bar{a}$.

(10 Marks)

(b) A system consists of three particles with the following masses and position vectors (in meters): $m_1 = 2 \text{ kg}$, $\bar{r}_1 = 2\mathbf{i} + 1\mathbf{j}$, $m_2 = 3 \text{ kg}$, $\bar{r}_2 = 1\mathbf{i} + 3\mathbf{j}$, $m_3 = 5 \text{ kg}$, $\bar{r}_3 = 4\mathbf{i} + 2\mathbf{j}$.

(i) Determine the position vector of the center of mass $\bar{r}_G$ of the system.

(ii) If the accelerations of the particles are: $\bar{a}_1 = (3\mathbf{i} + 0\mathbf{j}) \text{ m/s}^2$, $\bar{a}_2 = (2\mathbf{i} + 1\mathbf{j}) \text{ m/s}^2$, $\bar{a}_3 = (1\mathbf{i} + 2\mathbf{j}) \text{ m/s}^2$, compute the acceleration of the center of mass $\bar{a}_G$.

(iii) Use your result from (ii) to compute the resultant external force acting on the system, its magnitude and direction from the positive x axis.

(15 Marks)**Question Four****[25 Marks]**

(a) A smooth 2-kg collar C, shown in Figure Q. 4, is attached to a spring having a stiffness $k = 3 \text{ N/m}$ and an unstretched length of 0.75 m . If the collar is released from rest at A, determine its acceleration and the normal force of the rod on the collar at the instant $y = 1 \text{ m}$.

(15 Marks)

(b) Calculate the total work done by all the forces acting on the collar if it were to be restored to position A.

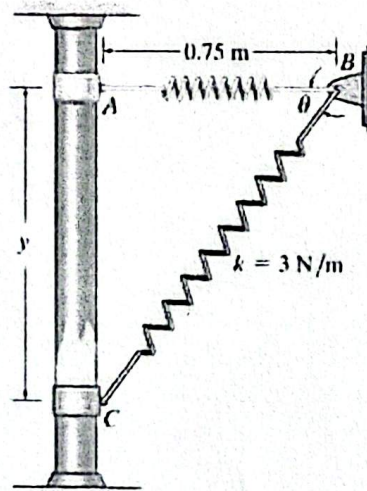
(10 Marks)

Figure Q. 4

$$N_c = 0.9 \text{ N}$$

$$a_y = 9.36 \text{ m/s}^2$$

Question Five**[25 Marks]**

(a) The disk shown in Figure Q. 5 is originally rotating at $\omega_0 = 8 \text{ rad/s}$. If it is subjected to a constant angular acceleration of $\alpha = 6 \text{ rad/s}^2$, determine the magnitudes of the velocity and the normal and tangential components of acceleration of point A at the instant $t = 0.5 \text{ s}$. **(10 Marks)**

(b) Referring to the disk in part (a), determine the magnitudes of the velocity and the normal and tangential components of acceleration of point B just after the wheel undergoes 2 revolutions. **(15 Marks)**

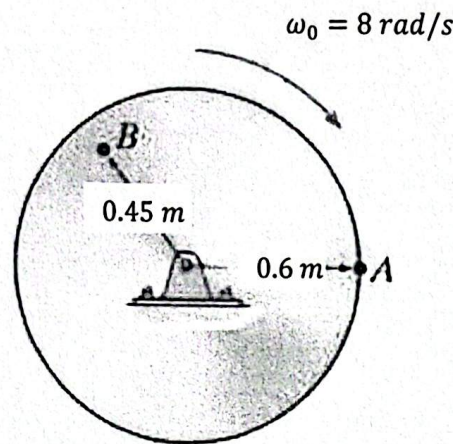


Figure Q. 5

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\omega^2 = 8^2 + 150.84$$